

Control and Performance of a Transformerless Cascade PWM STATCOM With Star Configuration

Hirofumi Akagi, *Fellow, IEEE*, Shigenori Inoue, *Member, IEEE*, and Tsurugi Yoshii

Abstract—This paper presents a three-phase transformerless cascade pulsewidth-modulation (PWM) STATic synchronous COMPensator (STATCOM) intended for installation on industrial and utility power distribution systems. It proposes a control algorithm that devotes itself not only to meeting the demand of reactive power but also to voltage balancing of multiple galvanically isolated and floating dc capacitors. The control algorithm based on a phase-shifted carrier modulation strategy is prominent in having no restriction on the cascade number. Experimental waveforms verify that a 200-V 10-kVA cascade PWM STATCOM with star configuration has the capability of inductive to capacitive (or capacitive to inductive) operation at the rated reactive power of 10 kVA within 20 ms while keeping the nine dc mean voltages controlled and balanced even during the transient state.

Index Terms—Cascade connection, multilevel converters, reactive power, STATic synchronous COMPensator (STATCOM), voltage balancing.

I. INTRODUCTION

A. Background

ATTENTION has been paid to a three-phase medium-voltage multilevel conversion system cascading multiple single-phase H-bridge converters in each phase [1]–[18]. This multilevel converter or inverter has been considered as an alternative to a three-level diode-clamped converter or inverter [19] for STATic synchronous COMPensator (STATCOM) and adjustable-speed motor drives.

When the multilevel converter is applied to STATCOM, each of the cascaded H-bridge converters should be equipped with a galvanically isolated and floating dc capacitor without any power source or circuit. This enables to eliminate a bulky, heavy, and costly line-frequency transformer from the cascade STATCOM. For example, the weight of a three-phase line-frequency transformer rated at 6.6 kV and 1 MVA ranges from 3000 to 4000 kg, while the weight of the three-phase cascaded converters with the same voltage and current ratings may range from 1000 to 2000 kg [20].

However, the cascade STATCOM suffers from a voltage imbalance between the multiple floating dc capacitors. Unequal conducting and switching losses produced by power switching

devices, as well as signal imbalance and resolution issues inherent in the control circuit including voltage/current sensors, may bring voltage imbalance to the dc capacitors in an actual system. The voltage imbalance is brought to an actual system by the following possible causes:

- tolerances of passive components;
- unequal conducting and switching losses produced by power switching devices;
- signal imbalance and resolution issues inherent in the control circuit including voltage/current sensors.

B. Classification of Cascade STATCOMs

Cascade STATCOMs can be classified into staircase modulation and pulsewidth modulation (PWM) in terms of voltage control. Research has been carried out in both staircase modulation and PWM, and the resultant papers have appeared in the technical literature [4]–[8], [18]. The authors of this paper prefer PWM to staircase modulation when a transformerless cascade STATCOM is being applied to 6.6-kV utility and industrial distribution systems in Japan. A main reason is that the 1.7-kV trench-gate insulated-gate bipolar transistors (IGBTs) can be operated at a switching frequency higher than 1 kHz with a low switching loss. Another reason comes from the availability of leading-edge digital signal processors (DSPs), field-programmable gate arrays (FPGAs), analog-to-digital converters, Hall-effect voltage/current sensors, and operational and isolation amplifiers at reasonable cost. As a result, PWM is superior in dynamic performance, more robust in line disturbances and faults, and more flexible in applications compared to staircase modulation.

The cascade PWM STATCOM for medium-voltage applications has attracted the attention of power electronics researchers/engineers who have been interested in power-factor correction and/or harmonic compensation [8], [11]–[15], [17], [18]. Nothing in the literature, however, has been presented or published on a practical control algorithm being capable not only of controlling and balancing all the dc mean voltages but also of expanding into a higher number of voltage levels.

C. Phase-Shifted Carrier Modulation Strategy

This paper deals with a 6.6-kV 1-MVA star-configured STATCOM cascading six single-phase H-bridge PWM converters per phase. It is followed by designing, constructing, and testing a 200-V 10-kVA star-configured STATCOM cascading three single-phase H-bridge PWM converters per phase. Giving priority to easy expansion into a high number of voltage levels

Paper IPCSD-07-015, presented at the 2006 Industry Applications Society Annual Meeting, Tampa, FL, October 8–12, and approved for publication in the IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS by the Industrial Power Converter Committee of the IEEE Industry Applications Society. Manuscript submitted for review December 11, 2006 and released for publication March 1, 2007.

The authors are with the Department of Electrical and Electronic Engineering, Tokyo Institute of Technology, Tokyo 152-8552, Japan (e-mail: akagi@ee.titech.ac.jp).

Digital Object Identifier 10.1109/TIA.2007.900487

leads to the use of a phase-shifted carrier modulation strategy in preference to phase disposition and centered space-vector modulation strategies [21], [22] with superior spectral performance. Thus, the PWM used in the STATCOM is the so-called “phase-shifted unipolar sinusoidal PWM” with a carrier frequency of 1 kHz.

The control algorithm considered in this paper devotes itself not only to meeting a demand of reactive power but also to controlling and balancing all the dc capacitor voltages. It is characterized by easy expansion into a high number of voltage levels. Experimental waveforms verify that the 200-V 10-kVA STATCOM can adjust the rated reactive power of 10 kVA from inductive to capacitive (or capacitive to inductive) within 20 ms while keeping the nine dc capacitor voltages controlled and balanced even during the transient state.

II. DESIGN OF THE 6.6-kV 1-MVA STATCOM

A. Cascade Number

For the sake of simplicity, the number of cascaded voltage-source H-bridge converters in each phase is referred to as a “cascade number” in this paper. When one tries to design the 6.6-kV STATCOM, the cascade number is one of the most important design parameters, which accounts for the blocking voltage of the IGBTs being used in the STATCOM. The 1.7-kV IGBTs are now available from the market at reasonable cost. When they are used for each H-bridge converter, the dc mean voltage should be designed to be around 1000 V so that the ac root mean square (rms) voltage of each H-bridge converter should be around 625 V.¹ This gives the cascade number N as

$$N = \frac{6600V}{\sqrt{3} \cdot 625V} = 6.1.$$

Therefore, the cascade number can be assigned as $N = 6$.

B. System Configuration

Fig. 1 shows the circuit configuration of the 6.6-kV 1-MVA star-configured STATCOM cascading six H-bridge PWM converters in each phase. All the IGBTs have the same voltage and current ratings as 1.7 kV and 200 A. The total number of IGBTs used for the 6.6-kV STATCOM is 72 ($= 4 \times 6 \times 3$). Each H-bridge converter has a floating dc capacitor with a maximal dc mean voltage of 1000 V. The so-called “phase-shifted unipolar sinusoidal PWM” with a carrier frequency of 1 kHz is applied to a cluster of six cascaded H-bridge converters in each phase. Therefore, the ac voltage of each cluster becomes a 13-level line-to-neutral PWM waveform with the lowest harmonic sideband centered at 12 kHz ($= 1 \text{ kHz} \times 2 \times 6$).

An ac inductor in each phase supports a difference between the utility ac voltage at the point of installation of the STATCOM and the 13-level PWM voltage. This inductor makes a significant contribution to filtering out switching rip-

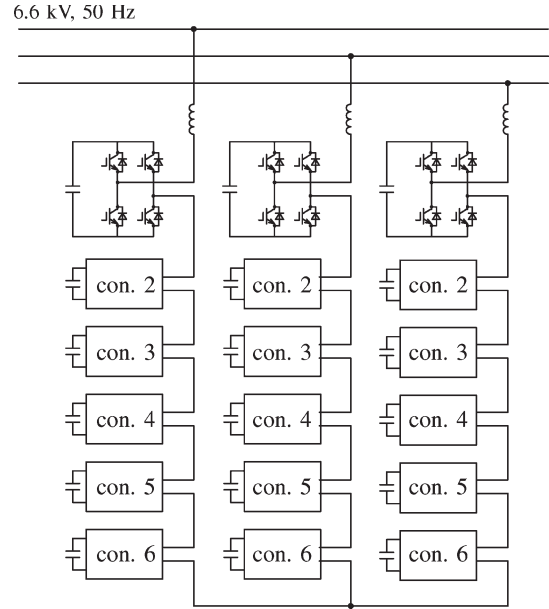


Fig. 1. Circuit configuration of the 6.6-kV 1-MVA star-configured cascade PWM STATCOM using 1.7-kV 200-A IGBTs.

ples caused by PWM, which thus results in an almost sinusoidal current without connecting any switching-ripple filter at the point of installation.

The following contribution is offered from the basic discussion of the 6.6-kV 1-MVA STATCOM: The control algorithm considered in this paper should have no restriction on the cascade number from a theoretical point of view. The reason is that such a high-power STATCOM is not confined to 6.6-kV applications in the world.

III. 200-V 10-kVA DOWNSCALED STATCOM

A. Cascade Number

A three-phase downscaled STATCOM rated at 200 V and 10 kVA is designed, constructed, and tested to verify the viability and effectiveness of the 6.6-kV STATCOM. It is important to assign an appropriate cascade number to the 200-V STATCOM. The cascade number of $N = 6$ is definitely the best choice in terms of exact downscale. However, the 200-V STATCOM with $N = 6$ is challenging to designing, constructing, and testing in the authors' laboratory. As described in Section V, the control algorithm has no restriction on the cascade number, so the authors assign $N = 3$ to the 200-V STATCOM, which verifies the capability of expanding the control algorithm into a higher number of voltage levels.

B. System Configuration

Fig. 2 shows the system configuration of the 200-V 10-kVA STATCOM. Table I summarizes the circuit parameters. It was scaled down from the 6.6-kV STATCOM in Fig. 1, except for assigning $N = 3$ to it. Thus, the 200-V STATCOM consists of nine H-bridge converters with the same voltage and current ratings. As a result, it produces a seven-level line-to-neutral (13-level line-to-line) voltage waveform. The phase-shifted

¹This ac voltage is given by $1000V/1.6$, where the number “1.6” in the denominator is a design parameter representing the ratio of the dc mean voltage to the ac rms voltage in a voltage-source H-bridge PWM converter.

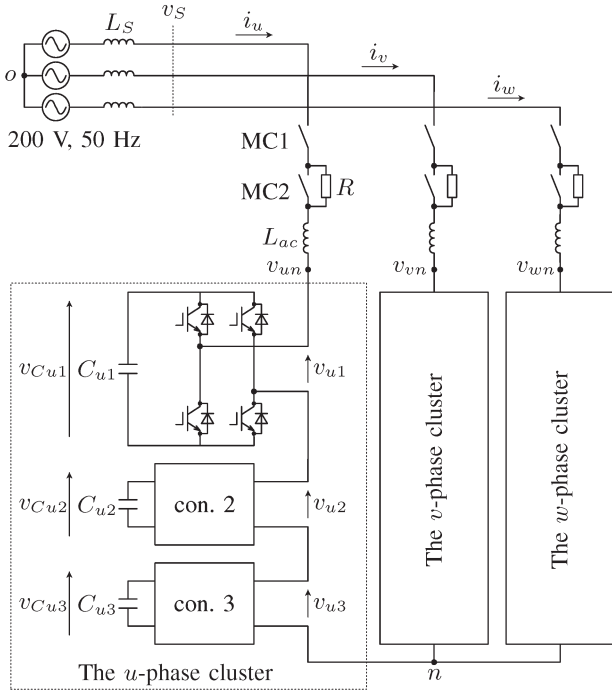


Fig. 2. 200-V 10-kVA downscaled STATCOM.

TABLE I
CIRCUIT PARAMETERS IN FIG. 2

Rated reactive power	Q	10 kVA
Nominal line-to-line rms voltage	V_S	200 V
Background system inductance	L_S	48 μ H (0.4%)
AC inductor	L_{ac}	1.2 mH (10%)
Starting resistor	R	10 Ω
DC capacitor voltage reference	v_C^*	60 ~ 70 V
DC capacitor capacitance	C	16,400 μ F
Unit capacitance constant	H	36 ms at 70 V
Carrier frequency for PWM		1 kHz
Low-pass-filter time constant	T	11 ms
Dead time in each converter		2 μ s

on a three-phase 200-V, 10-kVA, 50-kVA, 50-Hz base

unipolar sinusoidal PWM with a carrier frequency of 1 kHz is applied to the 200-V STATCOM. Each H-bridge converter is equipped with an isolated dc capacitor with a capacitance value of 16 400 μ F (the unit capacitance constant: $H = 36$ ms at 70 V [23]).² Note that neither external circuit nor power source, except for the dc capacitor and a Hall-effect voltage sensor, is installed on the dc side. For example, a cluster of three cascaded converters in the u -phase is referred to as the u -phase cluster in this paper. Moreover, v_{un} is the u -phase cluster voltage with respect to the neutral point of the star-configured STATCOM n , while v_{uv} is the u -phase cluster voltage with respect to the v -phase cluster voltage.

IV. CONTROL SYSTEM AND SAMPLING TIMING

Fig. 3 shows the control block diagram of the 200-V STATCOM. It consists of a fully digital control circuit based on

²This value of $H = 36$ ms is higher than that of a STATCOM based on a three-phase five-level diode-clamped converter [20]. An appropriate design of the unit capacitance constant is beyond the scope of this paper.

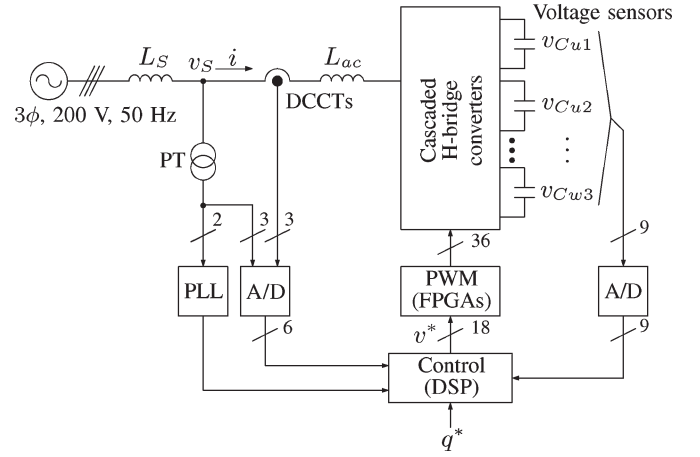
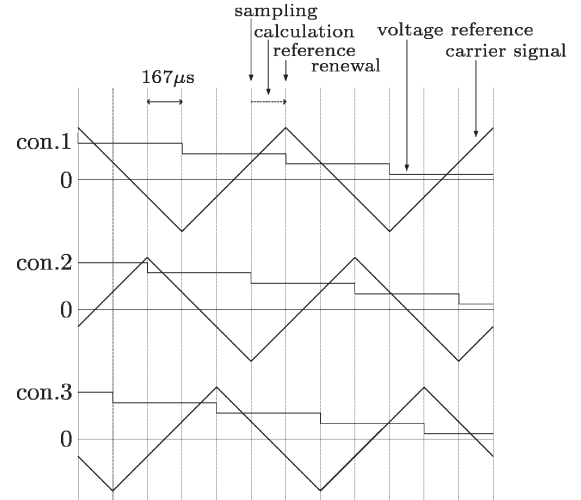


Fig. 3. Digital control system for the 200-V 10-kVA STATCOM.

Fig. 4. Three carrier signals with a phase shift of $2\pi/3$ between one and another along with sample timing.

a single DSP and three FPGAs. As shown in Fig. 4, three carrier signals with the same frequency as 1 kHz are phase-shifted by $2\pi/3$ from each other, which thus makes it possible to apply the phase-shifted carrier PWM to the three clusters. As a result, the line-to-neutral voltage at the ac side of the u -phase cluster v_{un} results in a seven-level PWM waveform with the lowest harmonic sideband centered at 6 kHz ($= 1 \text{ kHz} \times 2 \times 3$).

Data sampling for con. 1, for example, is carried out at every top or bottom of the carrier signal for con. 2. A pair of voltage references for con. 1 is renewed at the following top or bottom of the carrier signal for con. 1 with a sampling delay of 167 μ s ($= 1 \text{ ms}/6$). Then, it is held for 500 μ s to avoid the so-called “multiswitching.” As a result, the sampling period is 167 μ s (the sampling frequency is 6 kHz), and the reference renewal period is 500 μ s. The total delay time including the zero-order holder in the digital control is 667 μ s.

The digital control system in Fig. 3 has to execute a sequence of voltage/current signal acquisition and voltage-reference computation within the sampling period of 167 μ s. The DSP sends a pair of voltage references every 167 μ s to one of the three FPGAs that play an essential role in generating 36 PWM signals. A pair of voltage references corresponds to a

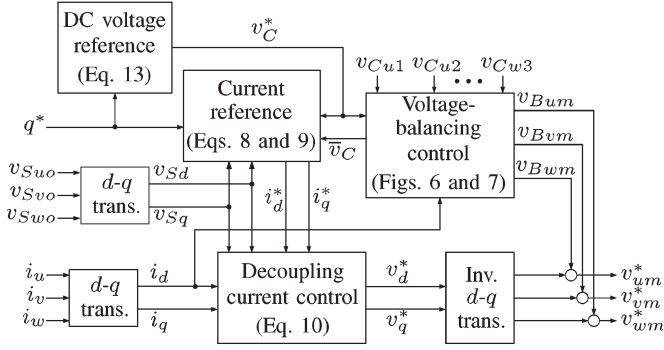


Fig. 5. Control block diagram for the 200-V 10-kVA STATCOM.

pair of legs in an H-bridge converter: One voltage reference is opposite in polarity to the other, which thus results in the so-called “unipolar modulation.” Note that one voltage reference is drawn while the other is eliminated from Fig. 4.

V. CONTROL ALGORITHM

Fig. 5 shows the block diagram of the control algorithm proposed in this paper. The whole control is divided mainly into the following three controls: reactive-power control and overall voltage control based on the d - q transformation, and voltage-balancing control for the nine dc capacitors. Moreover, the voltage-balancing control can be classified into “clustered balancing control” between the three clusters and “individual balancing control” between the three cascaded H-bridge converters in each cluster.

The phase-locked loop (PLL) circuit in Fig. 3 is used to synchronize internal control signals with the line phase ωt . As for the d - q and inverse d - q transformations in Fig. 5, the d -axis is aligned with the u -phase line-to-neutral voltage

$$v_{Suo} = \sqrt{\frac{2}{3}} V_S \sin \omega t. \quad (1)$$

This control algorithm is characterized by easy expansion into a STATCOM with a cascade number higher than four.

A. Reactive-Power Control and Overall Voltage Control

Neglecting resistance components from Fig. 2, the following set of voltage and current equations can be obtained:

$$\begin{bmatrix} v_{Suo} \\ v_{Svo} \\ v_{Swo} \end{bmatrix} - \begin{bmatrix} v_{un} \\ v_{vn} \\ v_{wn} \end{bmatrix} = L_{ac} \frac{d}{dt} \begin{bmatrix} i_u \\ i_v \\ i_w \end{bmatrix}. \quad (2)$$

Invoking the d - q transformation yields

$$\begin{bmatrix} L_{ac} \frac{d}{dt} & -\omega L_{ac} \\ \omega L_{ac} & L_{ac} \frac{d}{dt} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} v_{Sd} - v_d \\ v_{Sq} - v_q \end{bmatrix}. \quad (3)$$

Here, v_d and v_q are the d -axis and q -axis components corresponding to the three-phase cluster voltages v_{un} , v_{vn} , and v_{wn} ; and v_{Sd} and v_{Sq} are those corresponding to the three-phase supply voltages, respectively. When the three-phase supply

voltages v_{Suo} , v_{Svo} , and v_{Swo} are sinusoidal and balanced, v_{Sq} is always zero because v_{Suo} is aligned with the d -axis.

The instantaneous real power p and the instantaneous imaginary power q [24] are given as

$$p = v_{Sd} \cdot i_d + v_{Sq} \cdot i_q = v_{Sd} \cdot i_d \quad (4)$$

$$q = v_{Sd} \cdot i_q - v_{Sq} \cdot i_d = v_{Sd} \cdot i_q. \quad (5)$$

For example, the dc mean capacitor voltage in the u -phase cluster $\bar{v}_{Cu}$ can be calculated by

$$\bar{v}_{Cu} = \frac{1}{3} (\bar{v}_{Cu1} + \bar{v}_{Cu2} + \bar{v}_{Cu3}). \quad (6)$$

The dc mean capacitor voltage of the three clusters $\bar{v}_C$ is given by

$$\bar{v}_C = \frac{1}{3} (\bar{v}_{Cu} + \bar{v}_{Cv} + \bar{v}_{Cw}). \quad (7)$$

The d -axis current reference i_d^* and the q -axis current reference i_q^* are determined by

$$i_d^* = K_1 (v_C^* - \bar{v}_C) \quad (8)$$

$$i_q^* = \frac{q^*}{v_{Sd}}. \quad (9)$$

The d -axis voltage reference v_d^* and the q -axis voltage reference v_q^* are given by

$$\begin{bmatrix} v_d^* \\ v_q^* \end{bmatrix} = \frac{1}{3} \left(\begin{bmatrix} v_{Sd} \\ v_{Sq} \end{bmatrix} - \begin{bmatrix} 0 & -\omega L_{ac} \\ \omega L_{ac} & 0 \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} - K_2 \begin{bmatrix} i_d^* - i_d \\ i_q^* - i_q \end{bmatrix} - \frac{K_2}{T_2} \int \begin{bmatrix} 0 \\ i_q^* - i_q \end{bmatrix} dt \right). \quad (10)$$

The first and second terms on the right hand side of (10) are introduced to cancel out the supply voltage and the voltage appearing across the ac inductor. The third and fourth terms form a proportional controller for the d -axis current and a proportional plus integral controller for the q -axis current. The coefficient $1/3$ comes from the cascade number of $N = 3$. Note that the voltage control based on (8) and (10) considers a set of three clusters as a three-phase full-bridge converter using six IGBTs.

B. Clustered Balancing Control

Fig. 6 shows the block diagram of the clustered balancing control considering each of the three clusters as a single-phase H-bridge converter using four IGBTs. Attention is paid to the u -phase in the following. Each dc capacitor voltage detected by a Hall-effect voltage sensor contains a 100-Hz (double line frequency) component. A low-pass filter with a cutoff frequency of 15 Hz is used to eliminate the 100-Hz component from the detected dc voltage. This balancing control yields the u -phase clustered balancing signal v_{Bdu} from the common dc capacitor voltage reference v_C^* , the dc mean capacitor voltage of the u -phase cluster $\bar{v}_{Cu}$, and the d -axis current i_d as

$$v_{Bdu} = -K_4 \{ K_3 (v_C^* - \bar{v}_{Cu}) - i_d \} \sin \omega t. \quad (11)$$

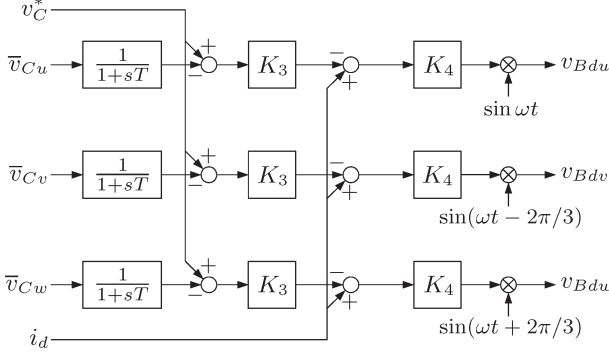


Fig. 6. Clustered balancing control between the u -phase, v -phase, and w -phase clusters considering each of the three clusters as a single-phase H-bridge converter using four IGBTs.

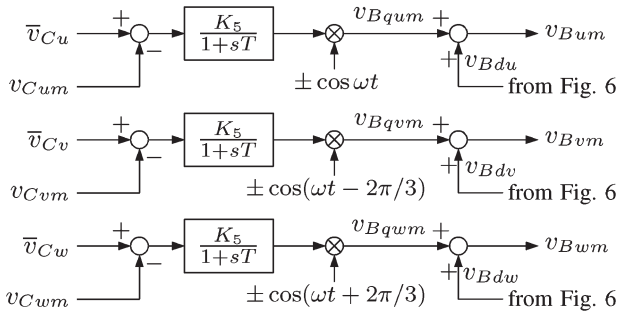


Fig. 7. Individual balancing control between the three cascaded converters inside each cluster, paying attention to the m th converter.

Note that v_C^* , $\bar{v}_{Cu}$, and i_d are not ac but dc signals. This makes it possible to form a minor current loop of i_d as if on the d -axis. Adding the current loop to the balancing control on the dc reference frame makes a significant contribution to improving stability. Multiplying the dc signal obtained from the minor current loop by “ $\sin \omega t$ ” implies transformation from the dc to the ac signal as if the inverse d - q transformation were applied. The signals “ $\sin(\omega t - 2\pi/3)$ ” and “ $\sin(\omega t + 2\pi/3)$ ” from the PLL circuit take the place of “ $\sin \omega t$ ” in v -phase and w -phase clusters.

C. Individual Balancing Control

Fig. 7 shows a block diagram of the individual balancing control. It plays an important role in balancing three dc mean capacitor voltages in each cluster. This balancing-control signal for the u -phase m th converter v_{Bqum} is given by

$$v_{Bqum} = \pm K_5(\bar{v}_{Cu} - \bar{v}_{Cum}) \cos \omega t. \quad (12)$$

Note that “ $\cos \omega t$ ” in (12) leads by 90° to the u -phase voltage given in (1). Making the sign of “ $\cos \omega t$ ” coincide with that of q^* yields an ac signal being in phase with a reactive current, which thus forms an amount of active power. Therefore, the individual balancing control does not work when $q^* = 0$. The following practical limitation is imposed on this experimental system:

$$1 \leq q^* \leq 10 \text{ [kVA]}.$$

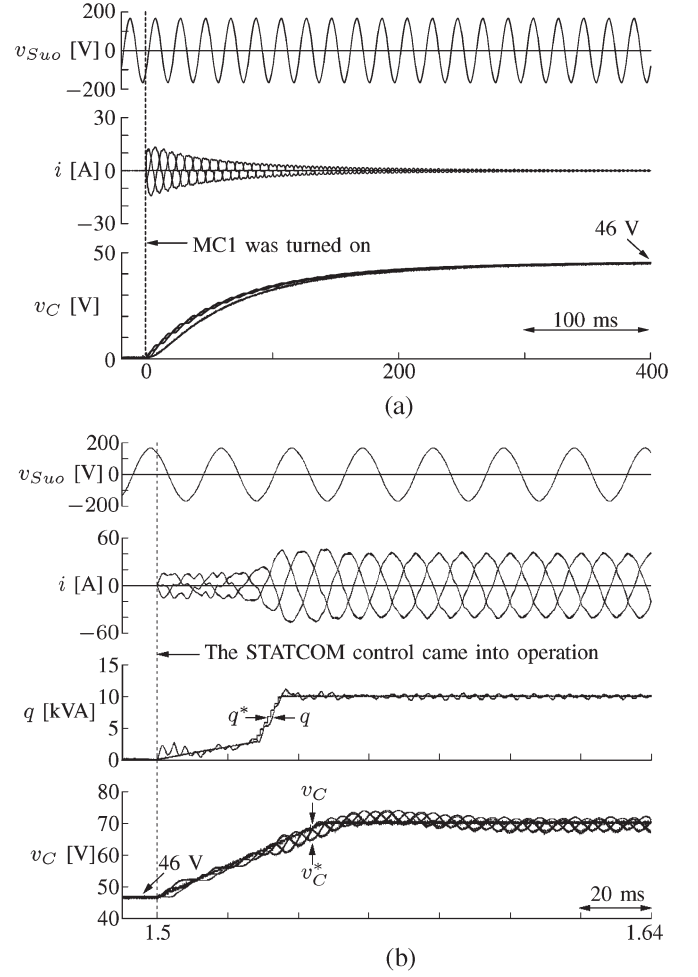


Fig. 8. Experimental waveforms when the STATCOM was started. (a) $-20 \text{ ms} \leq t \leq 400 \text{ ms}$. (b) $1.47 \text{ s} \leq t \leq 1.64 \text{ s}$.

The individual balancing control is intentionally designed to be slower in response speed than the clustered balancing control, as verified by experimental waveforms of v_C in Figs. 13 and 14. This can avoid a conflict of control between the two. However, it is not easy to assign appropriate values to gain parameters K_1 to K_5 for realizing the cascade PWM STATCOM with various voltage and current ratings. Due to page limitations, this paper makes no description of a design procedure for these gain parameters. The authors have a plan to make a detailed description of it in another paper.

VI. EXPERIMENTAL RESULTS

A. Startup Procedure

A specially designed startup circuit consists of two three-phase magnetic contactors MC1 and MC2, and a current-limiting resistor R in each phase, as shown in Fig. 2.

Fig. 8 shows experimental waveforms of the three-phase ac currents i_u , i_v , and i_w , and the nine dc capacitor voltages from v_{Cu1} to v_{Cw3} when the 200-V STATCOM was started. Unfortunately, it is difficult to distinguish the waveforms of the nine dc capacitor voltages. STATCOM takes the following startup procedure.

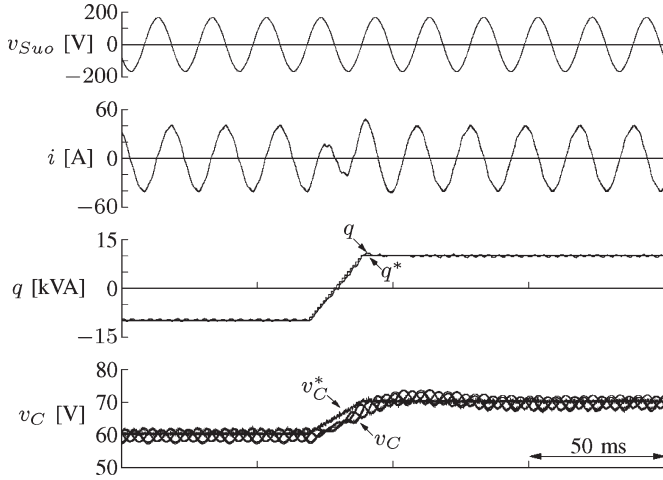


Fig. 9. Experimental waveforms in a transient state from inductive to capacitive operation at 10 kVA.

At time $t = 0$, MC1 was turned on, while MC2 remained switched off. An amount of ac current that is lower than the rated current of 30 A in each phase started to flow into the dc capacitors through a resistor of $R = 10 \Omega$. Thus, each dc capacitor voltage was charged up with a time constant of $R \times C/3 = 55$ ms. At time $t = 400$ ms, no current flowed, and the nine dc capacitors were equally charged up to a constant dc voltage of 46 V. At time $t = 1.0$ s, MC2 was turned on, but no current flowed, so no voltage change occurred in the nine dc capacitors. This phenomenon was not included in Fig. 8. At time $t = 1.5$ s, STATCOM control came into operation and started to draw a small amount of active power and a capacitive reactive power of 1 kVA from the three-phase utility grid, which thus controls the dc capacitor voltage reference v_C^* with a slow ramp function.³ This enables to gradually build up all the dc mean voltages from 46 V to its reference voltage of 60 V. Then, the reactive-power reference q^* was controlled with a ramp function from 1 to 10 kVA to prevent overvoltage from appearing across each dc capacitor.

B. Experimental Waveforms

Fig. 9 shows experimental waveforms in a transient state from inductive to capacitive operation with a ramp change in q^* from -10 to 10 kVA. Although v_C represents the nine dc capacitor voltages, it is difficult to distinguish the nine waveforms from Fig. 9. The voltage reference was controlled in a range between 60 and 70 V, which is being synchronized with the ramp change in q^* [kVA] as

$$v_C^* = 65 + q^*/2 \text{ [V]}. \quad (13)$$

The above equation means that capacitive operation at $q^* = 10$ kVA takes the maximal voltage reference of 70 V, and inductive operation at $q^* = -10$ kVA takes the minimal voltage reference of 60 V. Each dc voltage contains a 100-Hz

³Both clustered balancing control and individual balancing control were also put into operation at time $t = 1.5$ s, which thus keeps all nine dc capacitor voltages balanced well, as shown in Fig. 8(b).

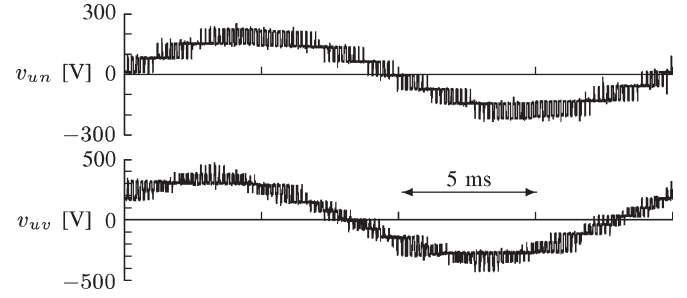


Fig. 10. Experimental waveforms when the STATCOM was put into capacitive operation at 10 kVA with $v_C^* = 70$ V.

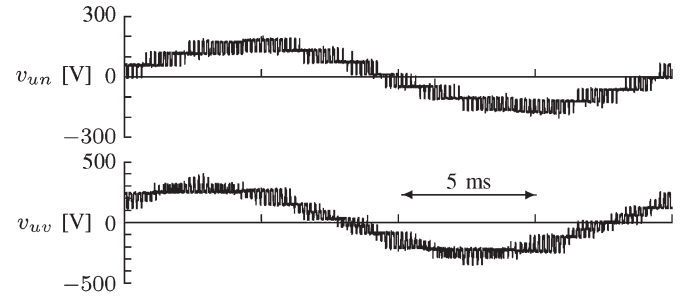


Fig. 11. Experimental waveforms when the STATCOM was put into inductive operation at 10 kVA with $v_C^* = 60$ V.

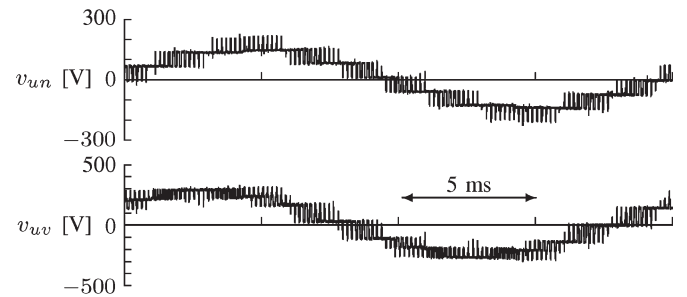


Fig. 12. Experimental waveforms when the STATCOM was put into inductive operation at 10 kVA with $v_C^* = 70$ V.

component inherent in a single-phase H-bridge converter. The 100-Hz component is superimposed on the dc mean voltage. Each dc mean voltage was kept balanced and controlled even in the transient state. The u -phase ac current flowing into the STATCOM (i_u) was slightly distorted with a current total harmonic distortion (THD) of 3.7%. The reason is discussed in the next section.

Fig. 10 shows experimental waveforms in capacitive operation at $q^* = 10$ kVA with $v_C^* = 70$ V, while Fig. 11 shows experimental waveforms in inductive operation at $q^* = -10$ kVA with $v_C^* = 60$ V. The u -phase cluster voltage with respect to the neutral point v_{un} is a seven-level waveform, and the u -phase cluster voltage with respect to the v -phase cluster voltage v_{uv} is a 13-level voltage waveform in both figures, as expected.

Fig. 12 shows experimental waveforms in an inductive operation of $q^* = -10$ kVA with $v_C^* = 70$ V. Note that v_{un} is a seven-level waveform, but v_{uv} looks like a nine-level voltage waveform. The reason is that Fig. 12 has a smaller modulation factor than Fig. 11.

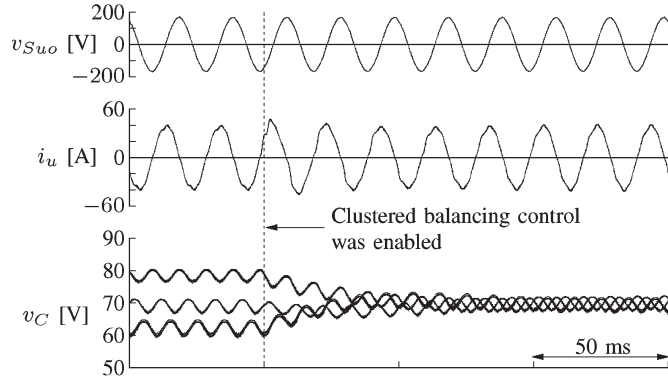


Fig. 13. Experimental waveforms confirming the effectiveness of the clustered balancing control when the STATCOM was put into capacitive operation at 10 kVA with $v_C^* = 70$ V. Both overall voltage control and individual balancing control remained active in this experiment.

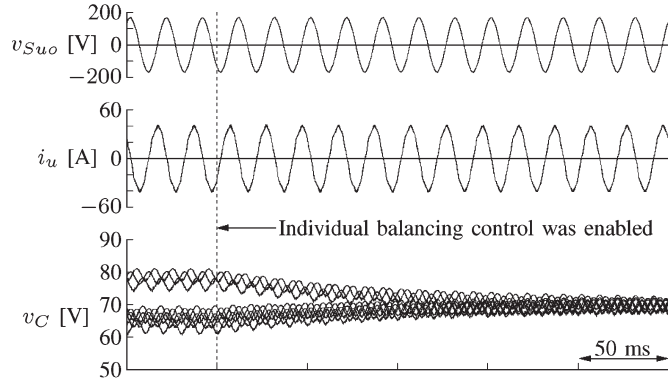


Fig. 14. Experimental waveforms confirming the effectiveness of the individual balancing control when the STATCOM was put into capacitive operation at 10 kVA with $v_C^* = 70$ V. Both overall voltage control and clustered balancing control remained active in this experiment.

Fig. 13 shows experimental waveforms of the u -phase ac current i_u and the nine dc capacitor voltages, which confirms the effectiveness of the clustered balancing control. During capacitive operation at 10 kVA with $v_C^* = 70$ V, the clustered balancing control was intentionally disabled for 3 s, and then it was enabled again. Both the overall voltage control based on (8) and (10), and the individual balancing control in Fig. 14 remained active through this experiment.

While the clustered balancing control was disabled, some waveforms of the nine dc capacitor voltages overlapped with others, which thus makes v_C look like three waveforms in Fig. 13. This means that three dc capacitor voltages in each cluster were well balanced because the individual balancing control remained active. As a result, voltage imbalance occurred between the three clusters. As soon as the clustered balancing control was enabled, each dc mean voltage of the nine dc capacitors started to converge at its reference voltage and finally reached 70 V in 70 ms. The main reason why the voltage imbalance occurred before enabling the clustered balancing control might be tolerances of electrolytic capacitors used as dc capacitors.

Fig. 14 shows experimental waveforms that confirm the effectiveness of the individual balancing control during capacitive operation at 10 kVA with $v_C^* = 70$ V. The individual

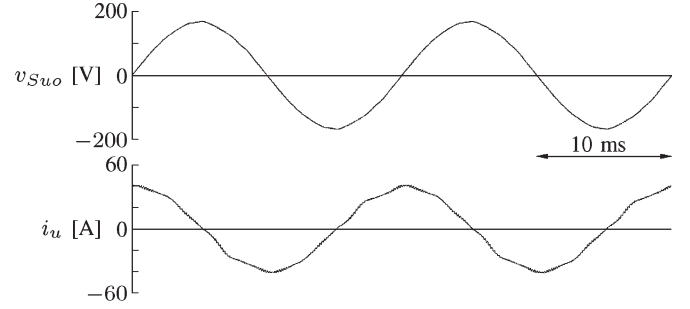


Fig. 15. Experimental waveforms when the STATCOM was put into capacitive operation at 10 kVA with $v_C^* = 70$ V.

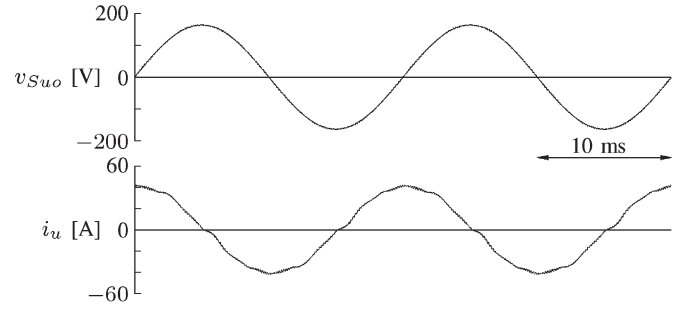


Fig. 16. Simulated waveforms when the STATCOM was put into capacitive operation at 10 kVA with $v_C^* = 70$ V, taking into account $V_{CE(Sat)} = V_F = 1.5$ V.

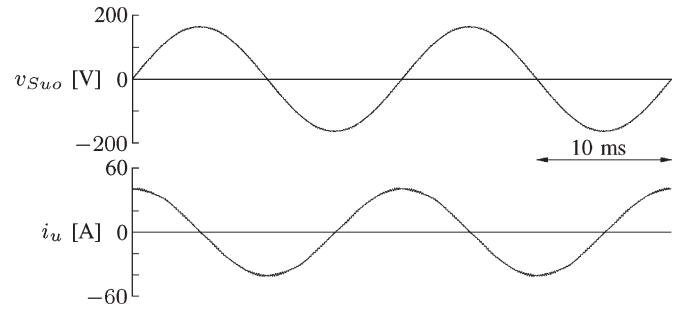


Fig. 17. Simulated waveforms when the STATCOM was put into capacitive operation at 10 kVA with $v_C^* = 70$ V, taking into account $V_{CE(Sat)} = V_F = 0.17$ V (= 0.25% of 70 V).

balancing control was intentionally disabled for 10 s, and then it was enabled again. Both overall voltage control and clustered balancing control remained active through this experiment. Whether the individual balancing control was disabled or enabled produced little effect on the waveform of i_u . However, voltage imbalance occurred in the nine dc capacitors, while the individual balancing function was disabled. As soon as it was enabled, each dc mean voltage of the nine dc capacitors started to converge at its reference voltage and finally reached 70 V in 150 ms.

Figs. 9, 13, and 14 conclude that the control algorithm proposed in this paper has the capability not only to control reactive power but also to balance the nine dc mean voltages.

VII. DISCUSSION OF CURRENT DISTORTION

Fig. 15 shows experimental waveforms of v_{Su0} and i_u during capacitive operation at 10 kVA. Note that both waveforms are time-expanded waveforms of v_{Su0} and i_u from Fig. 9 during

TABLE II
HARMONIC COMPONENTS AND THD OF AC CURRENT WITH CAPACITIVE OPERATION AT 10 kVA, EXPRESSED AS [%]

Harmonic order	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th	12th	13th	15th	THD
Experiment (Fig. 15)	1.1	1.4	0.5	1.9	0.3	2.3	0.5	0.6	0.4	0.8	0.1	0.5	0.0	3.7
Simulation I (Fig. 16)	0.0	0.0	0.0	2.0	0.0	2.7	0.1	0.0	0.1	1.6	0.0	1.1	0.0	3.9
Simulation II (Fig. 17)	0.0	0.1	0.0	0.7	0.0	0.6	0.0	0.0	0.0	0.4	0.0	0.3	0.0	1.1

Simulation I takes into account $V_{CE(Sat)} = V_F = 1.5$ V, and Simulation II takes into account $V_{CE(Sat)} = V_F = 0.17$ V (= 0.25% of 70 V).

the steady-state capacitive operation at 10 kVA. The waveform of i_u looks slightly distorted, so its THD reaches 3.7%.

The 36 IGBTs rated at 600 V and 150 A were used in this experiment, and each IGBT was integrated with a free-wheeling diode. The saturation voltage of the IGBT $V_{CE(Sat)}$ and the forward voltage of the diode V_F produce a bad effect on current-control performance because the dc mean voltage of each single-phase H-bridge converter is as low as 70 V.

Fig. 16 shows simulated waveforms of v_{Su} and i_u under the same operating conditions as in Fig. 15. This simulation assumed a pair of the IGBT and the diode as an ideal switch, except for taking into account a constant saturation/forward voltage of $V_{CE(Sat)} = V_F = 1.5$ V. In addition, it takes into account the following:

- a sampling delay of 167 μ s;
- a dead time of 2 μ s;
- a background system inductance of $L_S = 48$ μ H (0.4%).

These considerations enable to simulate more precise behavior in the digital controller and the nine single-phase H-bridge converters. As a result, the waveform of i_u in Fig. 16 is very similar to that in Fig. 15.

As described in Section II, the 6.6-kV STATCOM has the cascade number of $N = 6$, and each H-bridge converter consisting of four 1.7-kV IGBTs and a dc capacitor was designed to have a dc mean voltage of 1000 V. When the saturation/forward voltage of the 1.7-kV IGBT/diode is assumed as 2.5 V, it is only 0.25% of the dc mean voltage of 1000 V. This discussion indicates that the saturation/forward voltage corresponds to 0.17 V in a dc mean voltage of 70 V.

Fig. 17 shows simulated waveforms under the same operating conditions as in Fig. 16 except for taking into account the saturation/forward voltage of $V_{CE(Sat)} = V_F = 0.17$ V (= 0.25% of 70 V). The waveform of i_u is almost sinusoidal with much less distortion than that in Fig. 16.

Table II summarizes the harmonic components and THD of i_u with respect to the fundamental current. Their values were based on the waveforms obtained from Fig. 15 (experiment), Fig. 16 (simulation I), and Fig. 17 (simulation II). Each THD value counts up to the 40th harmonic current.

These considerations on Figs. 15–17, along with Table II, conclude that the distortion of i_u in Fig. 15 results from the saturation/forward voltage of the 600-V IGBT/diode. Therefore, the saturation/forward voltage produces little effect on the waveform of i_u in the actual 6.6-kV STATCOM.

VIII. CONCLUSION

This paper has addressed a transformerless cascade PWM STATCOM with star configuration for medium-voltage applications. The control algorithm is based on the so-called “phase-

shifted unipolar sinusoidal PWM” with a carrier frequency of 1 kHz. The control algorithm results from giving priority to the voltage-balancing control of multiple floating dc capacitors without restriction on the cascade number.

This leads to the 6.6-kV cascade PWM STATCOM using the 1.7-kV IGBTs that are now available from the market at reasonable cost. Experimental waveforms obtained from a 200-V 10-kVA downscaled STATCOM with a cascade number of $N = 3$ have verified the viability and effectiveness of the proposed control algorithm.

REFERENCES

- [1] T. Ohnishi, “Multiple single-phase PWM inverter by means of combination control,” *IEEJ Trans. Ind. Appl.*, vol. 115, no. 1, pp. 63–69, 1995. (in Japanese).
- [2] J. S. Lai and F. Z. Peng, “Multilevel converters—A new breed of power converters,” *IEEE Trans. Ind. Appl.*, vol. 32, no. 3, pp. 509–517, May/Jun. 1996.
- [3] P. Hammond, “A new approach to enhance power quality for medium-voltage ac drives,” *IEEE Trans. Ind. Appl.*, vol. 33, no. 1, pp. 202–208, Jan./Feb. 1997.
- [4] F. Z. Peng and J. S. Lai, “Dynamic performance and control of a static var generator using multilevel inverters,” *IEEE Trans. Ind. Appl.*, vol. 33, no. 3, pp. 748–755, May/Jun. 1998.
- [5] J. D. Ainsworth, M. Davies, P. J. Hitz, K. E. Owen, and D. R. Trainer, “Static var compensator (STATCOM) based on single-phase chain circuit converters,” *Proc. Inst. Electr. Eng.—Gener. Transm. Distrib.*, vol. 145, no. 4, pp. 381–386, Jul. 1998.
- [6] F. Z. Peng, J. W. McKeever, and D. J. Adams, “A power line conditioner using multilevel inverters for distribution systems,” *IEEE Trans. Ind. Appl.*, vol. 34, no. 6, pp. 1293–1298, Nov./Dec. 1998.
- [7] L. M. Tolbert, F. Z. Peng, and T. G. Habetler, “Multilevel converters for large electric drives,” *IEEE Trans. Ind. Appl.*, vol. 35, no. 1, pp. 36–44, Jan./Feb. 1999.
- [8] Y. Liang and C. O. Nwankpa, “A new type of STATCOM based on cascading voltage-source inverters with phase-shifted unipolar SPWM,” *IEEE Trans. Ind. Appl.*, vol. 35, no. 5, pp. 1118–1123, Sep./Oct. 1999.
- [9] M. D. Manjrekar, P. K. Steimer, and T. A. Lipo, “Hybrid multilevel power conversion system: A competitive solution for high-power applications,” *IEEE Trans. Ind. Appl.*, vol. 36, no. 3, pp. 834–841, May/Jun. 2000.
- [10] L. M. Tolbert, F. Z. Peng, and T. G. Habetler, “Multilevel PWM methods at low modulation indices,” *IEEE Trans. Power Electron.*, vol. 15, no. 4, pp. 719–725, Jul. 2000.
- [11] J. Rodrigues, J. S. Lai, and F. Z. Peng, “Multilevel inverters: A survey of topologies, controls, and applications,” *IEEE Trans. Ind. Electron.*, vol. 49, no. 4, pp. 724–738, Aug. 2002.
- [12] M. Lopez, L. Moran, J. Espinoza, and J. Dixon, “Performance analysis of a hybrid asymmetric multilevel inverter for high-voltage active power filter applications,” in *Proc. IEEE IECON*, 2003, pp. 1050–1055.
- [13] A. M. Massoud, S. J. Finney, and B. W. Williams, “Seven-level active power filter,” in *Proc. IEEE Int. Conf. Harmonics and Quality Power*, 2004, pp. 136–141.
- [14] L. A. Silva, S. P. Pimentel, and J. A. Pomilio, “Nineteen-level active filter system using asymmetrical cascaded converter with dc voltage control,” in *Proc. IEEE PESC*, 2005, pp. 303–308.
- [15] K. Fujii, U. Schwarzer, and R. W. De Doncker, “Comparison of hard-switched multi-level inverter topologies for STATCOM by loss-implemented simulation and cost estimation,” in *Proc. IEEE PESC*, 2005, pp. 340–346.
- [16] L. A. Silva, S. P. Pimentel, and J. A. Pomilio, “Analysis and proposal of capacitor voltage control for an asymmetrical cascaded inverter,” in *Proc. IEEE PESC*, 2005, pp. 809–815.

- [17] D. Soto and T. C. Green, "A dc link capacitor voltage control strategy of a PWM cascade STATCOM," in *Proc. IEEE PESC*, 2005, pp. 2251–2256.
- [18] K. Fujii, R. W. De Doncker, and S. Konishi, "A novel dc-link voltage control of PWM-switched cascade cell multi-level inverter applied to STATCOM," in *Conf. Rec. IEEE IAS Annu. Meeting*, 2005, pp. 961–967.
- [19] A. Nabae, I. Takahashi, and H. Akagi, "A new neutral-point-clamped PWM inverter," *IEEE Trans. Ind. Appl.*, vol. 1A-17, no. 5, pp. 518–523, Sep./Oct. 1981.
- [20] H. Akagi, H. Fujita, S. Yonetani, and Y. Kondo, "A 6.6-kV transformerless STATCOM based on a five-level diode-clamped PWM converter: System design and experimentation of a 200-V, 10-kVA laboratory model," in *Conf. Rec. IEEE IAS Annu. Meeting*, 2005, pp. 557–564.
- [21] B. P. McGrath and D. G. Holmes, "Multicarrier PWM strategies for multilevel inverters," *IEEE Trans. Ind. Electron.*, vol. 49, no. 4, pp. 858–867, Aug. 2002.
- [22] B. P. McGrath, D. G. Holmes, and T. A. Lipo, "Optimised space vector switching sequences for multilevel inverters," *IEEE Trans. Power Electron.*, vol. 18, no. 6, pp. 1293–1301, Nov. 2003.
- [23] H. Fujita, S. Tominaga, and H. Akagi, "Analysis and design of a dc voltage-controlled static var compensator using quad-series voltage-source inverters," *IEEE Trans. Ind. Appl.*, vol. 32, no. 4, pp. 970–978, Jul./Aug. 1996.
- [24] H. Akagi, Y. Kanazawa, and A. Nabae, "Instantaneous reactive power compensators comprising switching devices without energy storage components," *IEEE Trans. Ind. Appl.*, vol. 1A-20, no. 3, pp. 625–630, May/Jun. 1984.



Shigenori Inoue (S'02–M'07) was born in Fujimi, Saitama, Japan, in 1979. He received the B.S. and M.S. degrees from Tokyo Metropolitan University, Tokyo, Japan, in 2002 and 2004, respectively, and the Ph.D degree from the Tokyo Institute of Technology, Tokyo, Japan, in 2007.

He is a Research Fellow of the Japan Society for the Promotion of Science (JSPS). His research interests include medium-voltage power conversion systems, bidirectional isolated dc–dc converters, SiC/GaN-based power devices, and active filters.



Tsurugi Yoshii was born in Yokohama, Japan, on April 4, 1983. He received the B.S. degree in 2006 from Tokyo Institute of Technology, Tokyo, Japan, where he is currently working toward the M.S. degree.

His research interests include cascade multilevel converters.



Hirofumi Akagi (M'87–SM'94–F'96) was born in Okayama, Japan, in 1951. He received the B.S. degree in electrical engineering from Nagoya Institute of Technology, Nagoya, Japan, in 1974, and the M.S. and Ph.D. degrees in electrical engineering from Tokyo Institute of Technology, Tokyo, Japan, in 1976 and 1979, respectively.

In 1979, he was with Nagaoka University of Technology, Nagaoka, Japan, as an Assistant and then an Associate Professor in the Department of Electrical Engineering. In 1987, he was a Visiting Scientist at Massachusetts Institute of Technology MIT, Cambridge, for ten months. From 1991 to 1999, he was a Professor in the Department of Electrical Engineering, Okayama University, Okayama. From March to August 1996, he was a Visiting Professor at the University of Wisconsin, Madison, and then MIT. Since January 2000, he has been a Professor in the Department of Electrical and Electronic Engineering, Tokyo Institute of Technology. He has published 68 IEEE TRANSACTIONS papers and two invited PROCEEDINGS OF THE IEEE papers. He has made presentations many times as a keynote or invited speaker internationally. His research interests include power conversion systems, ac motor drives, high-frequency resonant inverters for induction heating and corona discharge treatment processes, and utility applications of power electronics such as active filters for power conditioning, self-commutated BTB systems, and FACTS devices.

Dr. Akagi is currently the President of the IEEE Power Electronics Society. He was elected as a Distinguished Lecturer of the IEEE Industry Applications and Power Electronics Societies for 1998–1999. He received two IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS Prize Paper Awards in 1991 and 2004, two IEEE TRANSACTIONS ON POWER ELECTRONICS Prize Paper Awards in 1999 and 2003, nine IEEE Industry Applications Society Committee Prize Paper Awards, the IEEE William E. Newell Power Electronics Award in 2001, and the IEEE Industry Applications Society Outstanding Achievement Award in 2004.